Application Layer Formula Sheet

1. Core Performance Metrics

Traffic Intensity (or Link Utilization)

Measures how "busy" a network link is. It's the ratio of the rate at which data arrives at the link to the maximum rate (capacity) of the link.

$$\text{Traffic Intensity} = \frac{\text{Average Data Rate}}{\text{Link Capacity (Rate)}}$$

Average Data Rate The average amount of traffic flowing over the link, measured in bits per second (e.g., Mbps). For web browsing, this is often calculated as (Avg. Request Rate) $\times$ (Avg. Object Size).
Link Capacity The maximum transmission rate of the link (e.g., 100 Mbps).
How to Use A value close to 0 means the link is idle. A value close to 1 (e.g., 0.99) means the link is highly utilized, which leads to significant queueing delays. A value > 1 is an unstable state where the queue grows indefinitely.

Transmission and Propagation Delay

Transmission Delay (D_{trans}) The time required to push all of a packet's bits onto a link.

$$D_{trans} = \frac{L}{R}$$

L: Packet/Object size (in bits).
R: Link transmission rate (in bits per second).
Propagation Delay (D_{prop}) The time it takes for one bit to travel across the physical medium.

$$D_{prop} = \frac{d}{s}$$

d: Length of the physical link (in meters).
s: Propagation speed of the medium (e.g., $\approx 2 \times 10^8$ m/s).

2. HTTP Performance

Response Time for Non-persistent HTTP (per object)

Calculates the total time to fetch one object when a new TCP connection must be established for it.

$$\text{Response Time} \approx 2 \cdot \text{RTT} + \text{File Transmission Time}$$

RTT Round-Trip Time. The time for a small packet to travel from client to server and back.
File Transmission Time The time to send all bits of the object, calculated as $\frac{\text{Object Size}}{\text{Link Rate}}$.

Explanation The formula accounts for:
- **1 RTT** for the TCP handshake (SYN/SYN-ACK).
- **1 RTT** for the HTTP request to be sent and the response to begin arriving.
- The time to transmit the actual file.

Queueing Delay (Simplified Model)

Estimates the time a packet waits in a router's buffer due to congestion.

$$D_{queue} \approx \left(\frac{\text{Traffic Intensity}}{1 - \text{Traffic Intensity}}\right) \times \left(\frac{\text{Avg. Object Size}}{\text{Link Rate}}\right)$$

How to Use This shows that as traffic intensity approaches 1, the queueing delay approaches infinity. It is a critical component of total end-to-end delay in congested networks.

3. File Distribution Time (Client-Server vs. P2P)

These formulas calculate the *minimum time* to distribute a file of size **F** from one server to **N** peers.

Client-Server Distribution Time

The time is determined by the bottleneck, which is either the server uploading all copies or the slowest client downloading its copy.

$$D_{c-s} \geq \max\left\{\frac{N \cdot F}{u_s}, \frac{F}{d_{min}}\right\}$$

F Size of the file (bits).
N Number of peers (clients).
u_s Upload rate of the server (bps).
d_{min} Download rate of the slowest peer (bps).
Explanation The term $\frac{N \cdot F}{u_s}$ represents the server bottleneck. The time grows **linearly** with the number of peers, N, so it does not scale well.

Peer-to-Peer (P2P) Distribution Time

The time is determined by the bottleneck among three factors: the server uploading one copy, the slowest client downloading, and the entire system's combined upload capacity.

$$D_{P2P} \geq \max\left\{\frac{F}{u_s}, \frac{F}{d_{min}}, \frac{N \cdot F}{u_s + \sum_{i=1}^{N} u_i}\right\}$$

u_i Upload rate of peer i (bps).
$\sum u_i$ The sum of the upload rates of all peers in the torrent.
Explanation The three terms represent:
1. The time for the server to inject one copy of the file into the system.
2. The time for the slowest peer to receive its copy.
3. The time for the entire system (server + all peers) to upload all N copies of the file.

This model is **self-scaling** because as N increases, the total upload capacity ($u_s + \sum u_i$) also increases, preventing the distribution time from growing linearly.